COLCHICINE

Narrow therapeutic index

Diarrhoea = toxicity sign

CHEMICAL NAME N-[(7S)-1,2,3,10-tetramethoxy-9-oxo-5,6,7,9-tetrahydrobenzo[a]heptalen-7-yl]acetamide

CLASS Antigout agent (anti-inflammatory alkaloid, microtubule inhibitor)

TRADE NAMES Colchicine (generic) · Colcrys · Colchicum Dispert

THERAPEUTIC USES / INDICATIONS

- Acute gout attack (start within 12–24 h).
- Prophylaxis of gout flares when starting urate-lowering therapy.
- Familial Mediterranean fever.
- Pericarditis (adjunct); Behçet's disease.

ADMINISTRATION

- **PO** tablets with or without food.
- No IV use (fatal toxicity).

DOSAGES

- **Acute gout:** 1 mg, then 0.5 mg 1 h later (max 1.5 mg on day 1), then 0.5 mg up to 2–3 times daily; do not repeat course within 3 days.
- **Prophylaxis:** 0.5 mg once or twice daily.
- **FMF:** 1–2 mg/day.
- Reduce dose in elderly and renal impairment; avoid if CrCl <10 mL/min.

PHARMACOKINETICS

- **Onset:** pain relief within 12–24 h.
- Absorbed in gut; metabolised by CYP3A4; P-glycoprotein substrate.
- Excreted in bile and urine. **t½:** 20–40 h.

PHARMACODYNAMICS

- Binds tubulin → inhibits microtubule polymerisation in neutrophils.
- Reduces neutrophil migration and inflammatory response to urate crystals.
- Does not lower uric acid.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea, vomiting, abdominal pain.
- Headache.

ADVERSE EFFECTS (SERIOUS)

- Toxicity: severe diarrhoea, bone-marrow suppression, multi-organ failure.
- Myopathy and neuropathy (especially with statins, renal impairment).
- Agranulocytosis, aplastic anaemia.
- Rhabdomyolysis.

CONTRAINDICATIONS & CAUTIONS

- Severe renal or hepatic impairment.
- Blood dyscrasias.
- Use with strong CYP3A4 or P-gp inhibitors in renal/hepatic impairment.
- Pregnancy (relative).

KEY INTERACTIONS

- **Clarithromycin, erythromycin, ketoconazole, verapamil, diltiazem, ciclosporin:** ↑ colchicine → fatal toxicity.
- **Statins, fibrates:** ↑ myopathy.
- **Digoxin:** ↑ myopathy risk.

NURSING MANAGEMENT

- Check renal function and all current drugs before giving.
- Stop and report diarrhoea, vomiting or abdominal pain (early toxicity).
- Monitor CBC in long-term use.
- Assess joint pain, swelling and redness.
- Watch for muscle weakness.

PATIENT EDUCATION

- Stop and seek advice if diarrhoea or vomiting starts.
- Do not exceed the prescribed dose; overdose can be fatal.
- Avoid grapefruit juice.
- Keep out of children's reach.

KEY POINTS

- Inhibits neutrophil microtubules; does not lower urate.
- Diarrhoea = sign of toxicity.
- Low-dose regimen: 1 mg then 0.5 mg.
- Dangerous with clarithromycin, verapamil.